(a) Each order is needed

learning-value gap $|E_P J - E_Q J|$

10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}

3 6 9 12

order N

scalar witness, Thm 4.4(c): lower moments equal

parity witness, Thm 4.4(c): all degree < N moments equal

$N!/(2^{N-1}N^{N})$

(b) Moment count is exact

18/18 equal
(exact integer rank)

rank of monomial matrix on grid

10^{0} 10^{1}

10^{0} 10^{1}

C(k+N, N)

k=1 k=4
k=2 y = x
k=3

(c) Fewer than $2^{k}-1$ features: error 1

worst-pair discrepancy

1.00
0.75
0.50
0.25
0.00

0 4 8 12

number of nonconstant features d

k=1 k=4
k=2 full interface (C9 assertion, PASS)
k=3

Exact finite computations.